TD2

Exercise 1

An IoT system uses an ESP32, a DHT22 temperature/humidity sensor, an LED, and a Wi-Fi connection.

1. Identify the role of each component.
2. Classify each component as input, processing, output, or communication.
3. Draw a block diagram showing how the components interact.

Exercise 2

Complete the following table:

Sensor	What does it measure or detect?	Output type
DHT22	?	?
LDR	?	?
PIR	?	?
HC-SR04	?	?
MQ-2	?	?

Exercise 3

An IoT device uses a 3000 mAh battery. During each minute:

- The sensor consumes 10 mA for 2 seconds.
- The microcontroller consumes 20 mA for 3 seconds.
- The device consumes 0.1 mA during the remaining time.

Assume the three operating periods do not overlap.

1. Calculate the average current consumption.
2. Estimate the battery lifetime in hours and days.
3. Give two ways to increase battery lifetime.

Exercise 4

A potentiometer is connected to an ESP32 ADC input. Its output voltage varies from 0 to 3.3 V.

1. Is the potentiometer an analog or digital sensor?
2. What is the role of the ADC?
3. For a 12-bit ADC, how many different values can be represented?
4. What digital value approximately corresponds to an input voltage of 1.65 V, assuming an ideal 0–3.3 V range?

Exercise 5

Consider three IoT networks:

- Network A: All sensors communicate directly with one central gateway.
- Network B: Sensors can communicate with neighboring sensors to forward data.
- Network C: Two devices communicate directly with each other.

1. Identify the topology used in each network.
2. Which topology offers alternative communication paths?
3. What happens if the central gateway in Network A fails?

Exercise 6

A connected door lock uses an ESP32, an RFID reader, RFID cards, and an electronic lock.

1. Identify the hardware components.
2. Identify three software functions required by the system.
3. Explain how the system verifies whether a user is authorized.
4. Why is a relay or suitable driver needed to control some electronic locks?

Exercise 7

A parking lot contains 20 parking spaces. The objective is to detect available spaces and display their status on a web dashboard.

1. Which sensors could be used to detect vehicle presence?
2. Identify the processing and communication components.
3. Draw the system architecture.
4. What should the dashboard display if a sensor stops working?
5. Suggest two ways to reduce energy consumption.

Exercise 8

Choose an appropriate actuator or control component for each application and justify your answer.

1. Opening and closing an irrigation water valve.
2. Controlling the speed of a DC motor.
3. Switching a water pump ON and OFF.
4. Moving a robotic arm to a specific angle.
5. Adjusting the brightness of an LED.